

1.

Let Q denote a queue containing 4 numbers i.e. [11,22,33,44] and S be an empty stack. $\text{Head}(Q)$ returns the element at the head of the queue Q without removing it from Q . Similarly $\text{top}(S)$ returns the elements at the top of S without removing it from S . Consider the algorithm below:

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if  $S$  is empty or  $\text{top}(S) \leq \text{head}(Q)$  then  
     $x = \text{dequeue}(Q)$   
     $\text{push}(S, x)$   
else  
     $x = \text{pop}(S)$   
     $\text{enqueue}(Q, x)$   
end
```

Which element will be at the front of the queue after execution of this code two times.

- A. 11
- B. 22
- C. 33
- D. 44

Answer: C

2.

_____ are often used in multi-threaded environments as a safe way to exchange data between threads.

- A. Stacks
- B. Queue
- C. Sorting
- D. Searching

Answer: C

3.

A queue is implemented using a linear singly linked list. The queue has a head pointer and a tail pointer. Let n denote the number of nodes in the queue. Let enqueue be implemented by inserting a new node at the head, and dequeue be implemented by deletion of a node from the tail. Which one of the following is the time complexity of the most time-efficient implementation of enqueue and dequeue operations respectively, for this data structure?

- A. $\theta(1)$, $\theta(1)$
- B. $\theta(1)$, $\theta(n)$
- C. $\theta(n)$, $\theta(1)$
- D. $\theta(n)$, $\theta(n)$

Answer: B

4.

Consider an array based implementation of a circular queue where the front and rear has been initialized to -1 and Array size is N .

Which of the following statements is true for the number of elements or size of the queue?

S1: if $\text{front} > \text{rear}$ then $\text{size} = N - \text{front} + \text{rear} + 1$

S2: If $\text{front} \leq \text{rear}$ then $\text{size} = \text{rear} - \text{front} + 1$

- A. S1 is true but S2 is false
- B. S1 is false but S2 is true
- C. Both statements are true
- D. Both statements are false

Answer: A

5.

A circular queue has been implemented using a singly linked list where each node consists of a value and a single pointer pointing to the next node. We maintain exactly two external pointers FRONT and REAR pointing to the front node and the rear node of the queue, respectively. Which of the following statements is/are CORRECT for such a circular queue, so that insertion and deletion operations can be performed in constant time $O(1)$ time?

- I. Next pointer of the front node points to the rear node.
- II. Next pointer of the rear node points to the front node.

- A. (I) only
- B. (II) only.
- C. Both (I) and (II).
- D. Neither (I) nor (II).

Answer: B